Navigation with IMU and Magnetometer

Introduction

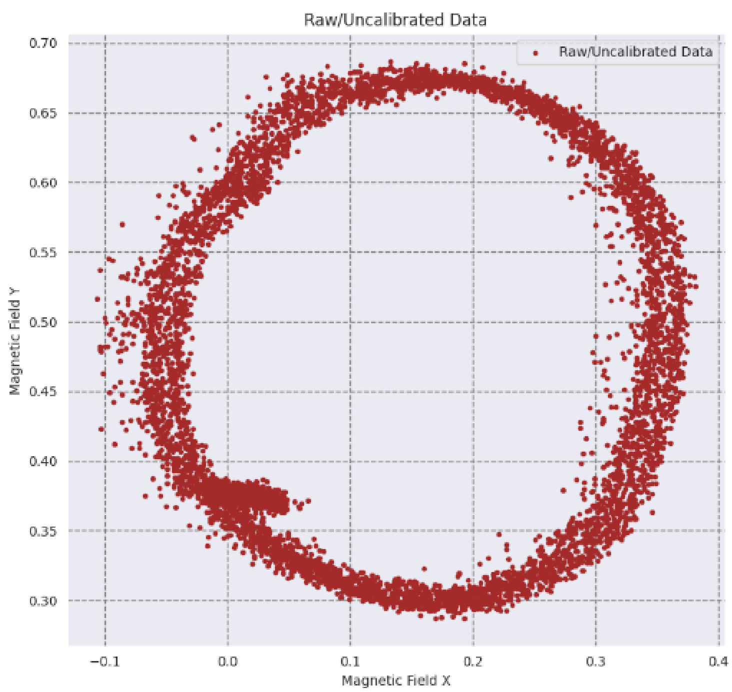
The report undertakes a comprehensive analysis of data from the Inertial Measurement Unit (IMU) and Global Positioning System (GPS) sensors, vital in advanced navigation and positioning systems. The IMU, particularly the VN 100 model by VectorNav Technologies, is a sophisticated module combining gyroscopes, accelerometers, and magnetometers for detailed motion tracking. Its sensor fusion algorithm is key to ensuring accuracy in dynamic environments. Meanwhile, GPS provides essential location data, offering positional (Northing and Easting) and altitude readings crucial in diverse applications from navigation to mapping. We focus on calibrating the magnetometer to correct distortions and on merging sensor data for enhanced yaw angle precision. Also, we assess techniques for forward velocity estimation and dead reckoning. This study aims to improve the precision and dependability of navigation systems, key to advancing autonomous and assisted driving technologies.

Data Collection

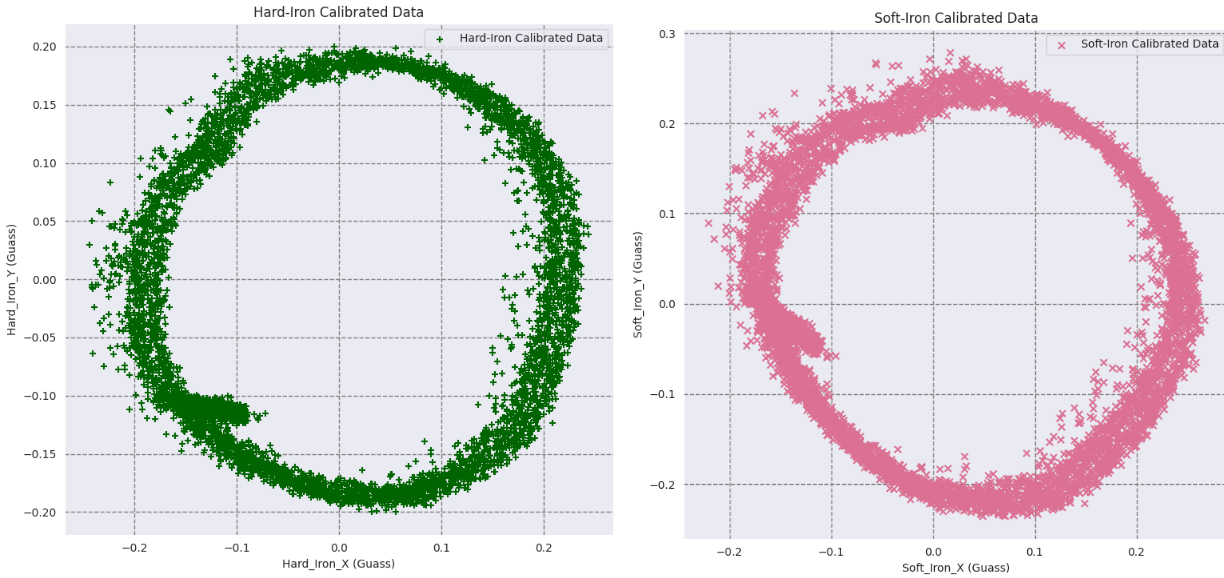
The data for this analysis was collected in a single rosbag file, incorporating readings from both IMU and GPS sensors. The first dataset, labeled as 'going in circles,' was recorded near Ruggles Station to capture circular driving patterns. The second set of data, representing diverse driving conditions, was gathered while navigating around Northeastern University. This methodical approach ensured a varied dataset, essential for a thorough evaluation of sensor capabilities in different driving scenarios.

Analysis

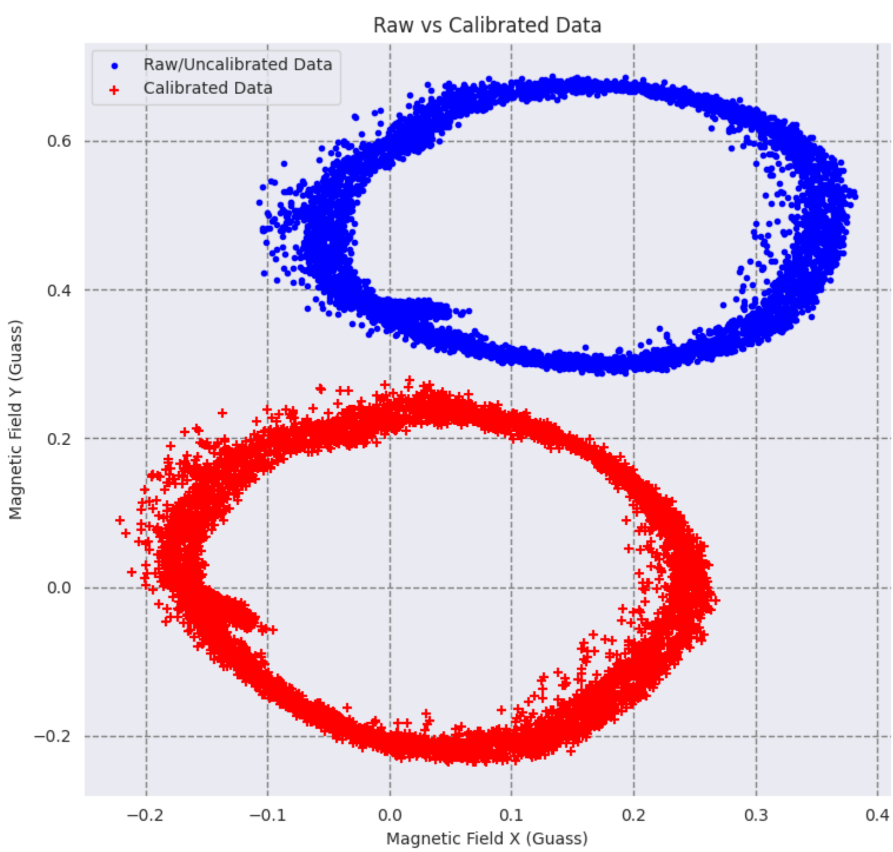
a. Magnetometer Calibration



In the magnetometer calibration process, I addressed hard-iron and soft-iron distortions using a two-step method. The first step involved calculating hard-iron offsets by identifying and subtracting the average of the maximum and minimum magnetic field readings on both the x and y axes, derived as 0.13745 and 0.4869, respectively. This correction is vital to eliminate the constant bias in the magnetic field measurements, evident from the data's shift from the origin. Next, the soft-iron calibration was more intricate, involving a covariance matrix of the adjusted data to identify the principal axes of the distortion ellipse. Eigenvalues and eigenvectors from this matrix were used to rotate the data, aligning it with these principal axes. The data was then scaled to correct for the elliptical distortion, normalizing it into a circle.

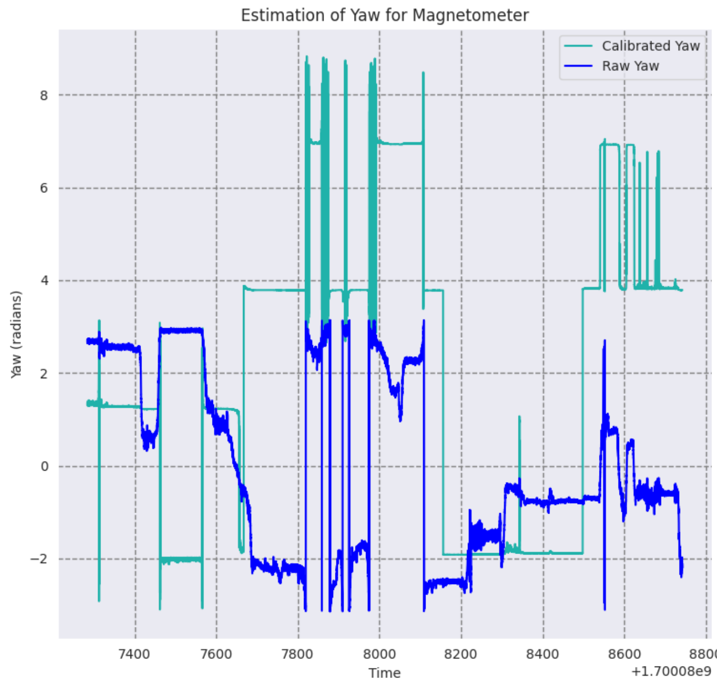


The initial data displayed these distortions clearly, with an elliptical shape and offset from the origin, both rectified after calibration. This detailed calibration process was critical for ensuring the magnetometer data's accuracy and reliability for precise navigation. The calibrated data, represented in circular patterns centered at the origin, now accurately reflects the Earth's magnetic field, crucial for navigation applications.

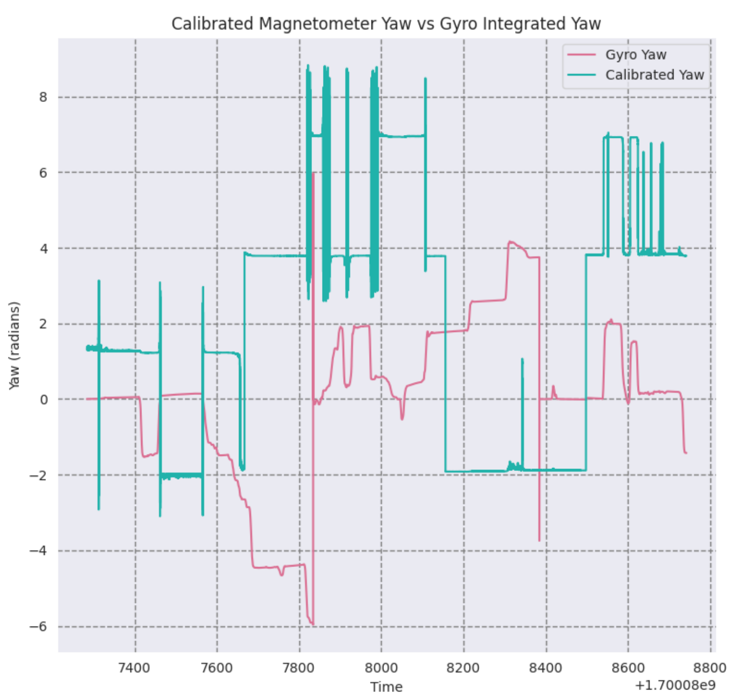


b. Sensor Fusion

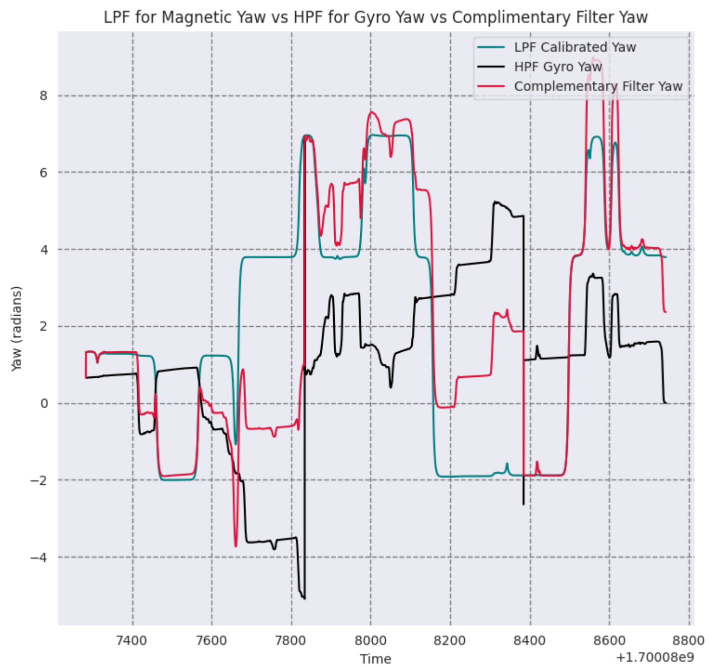
In the analysis, the magnetometer data was first calibrated to correct for hard and soft iron distortions. The hard iron calibration involved offsetting the data based on the maximum and minimum magnetic field readings. The soft iron calibration then normalized the data, transforming elliptical distortions into a circular pattern using eigenvector decomposition and scaling techniques. After calibration, the yaw angle was calculated from this data.



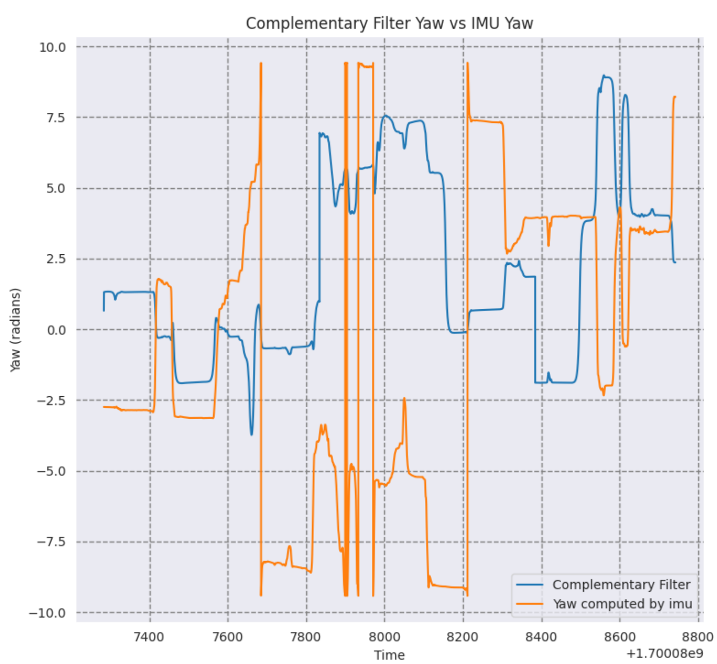
Concurrently, the gyro sensor data was integrated to estimate yaw angle, using a method that treats time series as a function to be integrated.



The calibrated magnetometer data and the integrated gyro data were then merged using a complementary filter. This filter combined a low-pass filtered magnetometer data (to minimize high-frequency noise) and a high-pass filtered gyro data (to reduce slow drift). The filter settings included a cutoff frequency of 0.05 Hz for the low-pass filter and 0.0001 Hz for the high-pass filter, optimizing the balance between long-term stability and short-term accuracy. This complementary approach provided a more reliable and comprehensive yaw estimation compared to using either sensor's data independently.

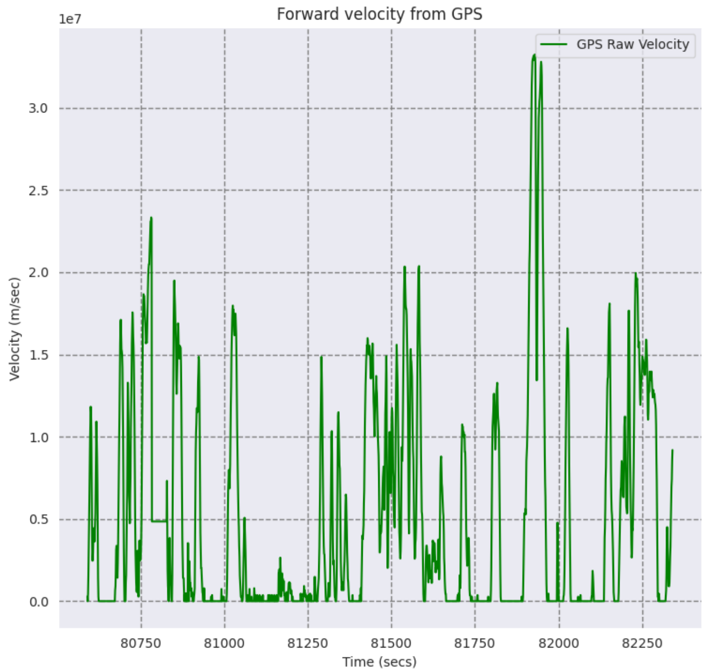


The analysis included comparing the yaw angle calculated from the IMU's internal algorithms with the estimate from the complementary filter. This comparison was pivotal in assessing the accuracy of the custom sensor fusion technique versus the IMU's sophisticated, integrated approach to yaw calculation, thereby offering insights into the efficacy of different methodologies in navigation applications.

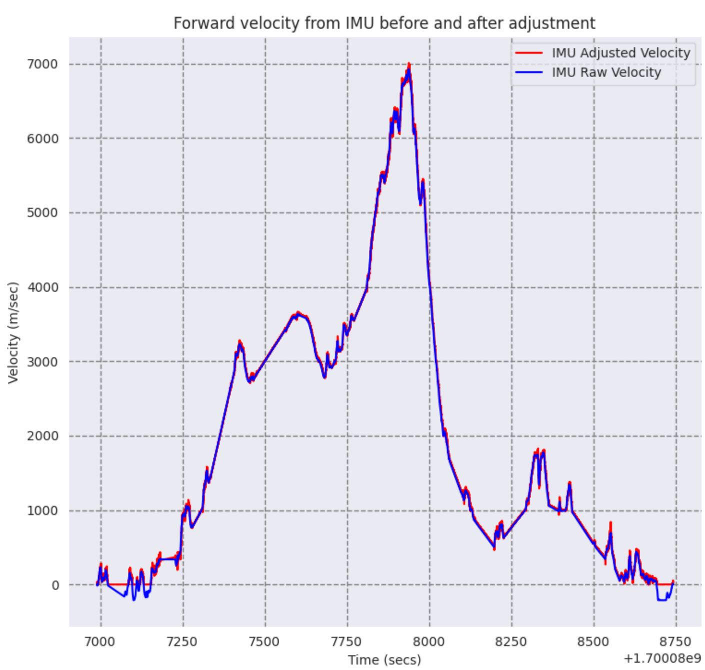


c. Forward Velocity

In this analysis, the code integrates acceleration data from the IMU to estimate velocity and compares it with GPS-derived velocity. The comparison between these two estimates highlights discrepancies possibly due to IMU noise and integration drift, versus GPS's positional change sensitivity. This analysis is crucial for understanding the strengths and limitations of each method in accurately capturing vehicle speed, vital for navigation and control systems in autonomous vehicles.



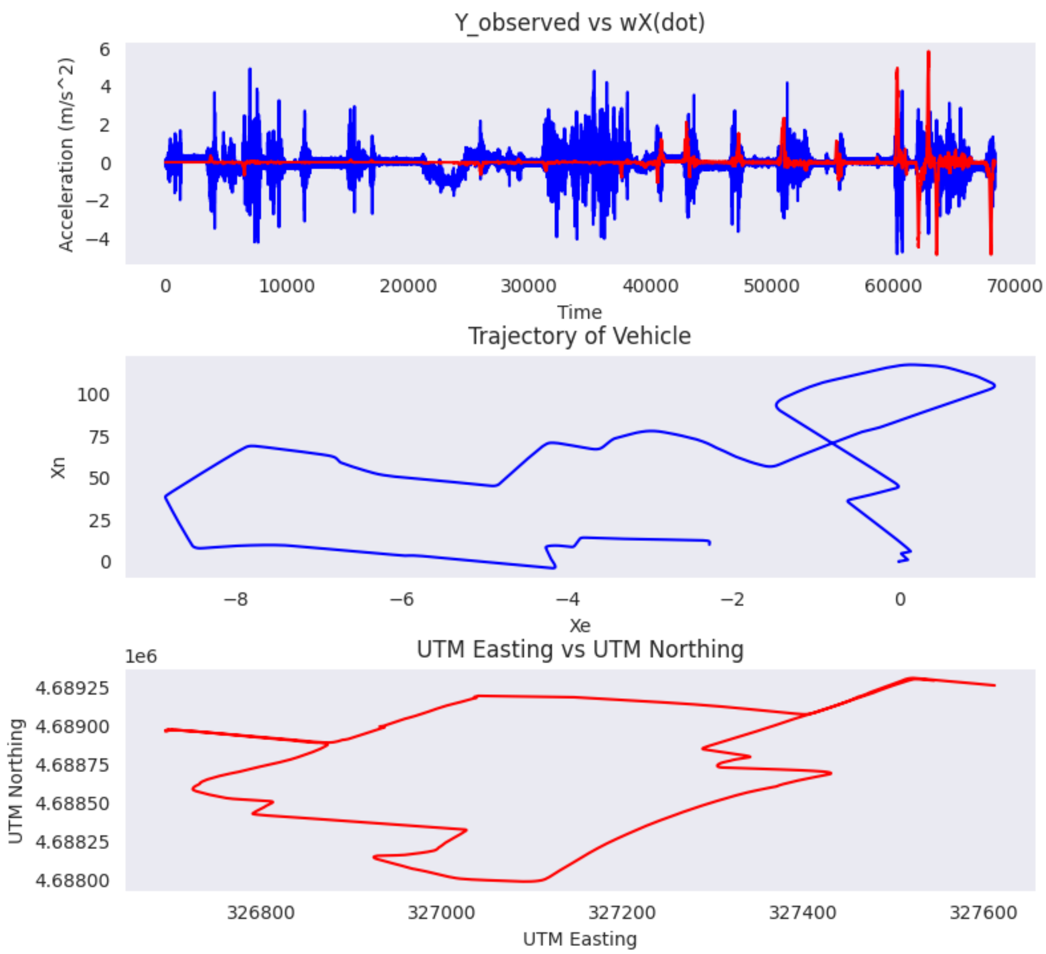
This plot is a direct measure of velocity, being calculated from positional changes over time. The GPS velocity is expected to be more stable and consistent but has its own set of errors like atmospheric delays or multipath effects.



The second plot illustrates the velocity estimate from the IMU accelerometer, both before and after adjustments. Initially, the accelerometer data required integration over time, which introduced cumulative errors, due to vehicle motion and axis velocity, leading to drift. Post-adjustment, the data was corrected for these errors, providing a more accurate representation of velocity. This plot is key in understanding the inherent challenges in using accelerometer data for velocity estimation and the effectiveness of the applied corrections.

d. Dead Reckoning

Dead reckoning in this analysis involved integrating forward velocity data, obtained from both GPS and the IMU, to estimate vehicle displacement. The process simplified the vehicle's motion to a two-dimensional plane, considering its position and rotation about the center of mass. The analysis compared these displacement estimates against GPS data to assess their accuracy. The integration of velocity data from the IMU, corrected for biases and errors, offered insights into the vehicle's trajectory, highlighting the efficacy and challenges of dead reckoning using IMU data in dynamic driving environments. The first plot compares the observed lateral acceleration from the IMU with the product of angular velocity and longitudinal velocity. It highlights the relationship and discrepancies between these two measurements, crucial for understanding the vehicle's lateral dynamics. The vehicle's trajectory plot, derived from integrated IMU data, visually represents the path taken by the vehicle. This plot is vital for assessing the accuracy of the dead reckoning process, especially when compared with GPS data. The final plot displays the vehicle's positional changes in UTM coordinates, showcasing the vehicle's movement in easting and northing dimensions. It serves as a benchmark for evaluating the accuracy of the trajectory estimated from the IMU data.



Discussion

1. Magnetometer Calibration

    - The calibration involved correcting for hard-iron (constant bias) and soft-iron (elliptical distortion) effects. Hard-iron effects were addressed by calculating and subtracting offsets, while soft-iron distortions were normalized using eigenvectors and covariance matrices. The distortions were evident from the elliptical pattern and shift from the origin in the uncalibrated data. These were corrected to a centered circular pattern post-calibration.

2. Complementary Filter for Yaw

    - A complementary filter merged low-pass filtered magnetometer data (cutoff at 0.05 Hz) with high-pass filtered gyro data (cutoff at 0.0001 Hz). Here 'filtfilt' and 'butter' is used to apply the filters. This balanced the long-term stability of the magnetometer with the short-term precision of the gyro. Therefore, minimized high-frequency noise in magnetometer data and reduced gyro drift.

3. Yaw Estimate for Navigation

    - The complementary filter output is most reliable due to its balanced approach, effectively integrating the short-term precision of gyro data with the long-term stability of magnetometer data. Therefore, providing a balanced and comprehensive yaw estimation.

4. Forward Velocity Adjustments

    - The adjustments were made to IMU data for biases, and the accelerometer data was integrated to estimate forward velocity. These adjustments helped align the IMU data with the actual vehicle motion.

5. Velocity Estimate Discrepancies

    - The discrepancies between accelerometer and GPS velocity estimates arose from cumulative errors in accelerometer data integration versus the direct measurement method of GPS.

6. Comparison of $\omega\dot{X}$ and $\ddot{y}_{obs}$

    - This equation represents the relationship between the observed linear acceleration ($\ddot{y}_{obs}$) of a point on a rigid body and the components of acceleration that contribute to it. The equation considers the linear acceleration of the center of mass ($\ddot{Y}$), the centripetal acceleration due to rotation ($\omega\dot{X}$), and the tangential acceleration due to the change in angular velocity ($\dot{\omega}x_c$). It is possible to relate the centripetal acceleration to the linear acceleration, and comparing the two can help determine the motion characteristics of the body. But there might be difference if the rigid body is not only rotating but also translating through space with a linear acceleration or if the angular velocity (ω) of the rotating body is changing over time then tangential acceleration ($\dot{\omega}x_c$) will also contribute to the difference between $\ddot{y}_{obs}$ and $\omega\dot{X}$ or it could be errors in the sensors could contribute to the difference between the two values.

7. Vehicle Trajectory Estimation

    - The vehicle trajectory, derived from IMU data, was compared with GPS data. Adjustments in headings and scaling (0.75 times) were made for accurate comparison, revealing a similar figure-eight pattern despite some positional errors. Scaling factors were used to ensure comparability between the tracks.

8. Navigation without Position Fix

    - Based on IMU performance and Allan deviation analysis, reliable navigation without a position fix is estimated at around 17-18 seconds. The IMU and GPS data matched closely for about 15 minutes, showing the efficacy of dead reckoning.

9. Estimation of

    - The position and velocity of the center-of-mass (CM) of the vehicle is shown by R and V, respectively. The inertial sensor is displaced from the CM by $r = (x_c , 0, 0)$ note that this vector is constant in the vehicle frame and assumes that the displacement of the IMU sensor is only along the x-axis. The velocity of the inertial sensor is: $v = V + \omega \times r$ ω is the is the z-axis angular velocity from the gyroscope $r = (x c , 0, 0)$ V is actual linear velocity v stands for measured linear velocity $\ddot{x} = \dot{v} + \omega \times v = \ddot{X} + \dot{\omega} \times r + \omega \times \dot{X} + \omega \times (\omega \times r)$ Then the range of possible x_c values were found using the equation: $x_c = (V - v)/\omega$. Using the above equation, we come to know that x_c and ω are inversely proportional, which means, if we take small values of ω it will give large value of x_c and vice versa. This method may isolate the radius of rotation of the car making the turn. And, when the car was driving in a straight line at a constant speed V was set as the minimum velocity, which would return a range of more reasonable estimates. X_c = (y_observed - xdot * omega)/ omegadot. The offset is determined by taking the mean of xc -> 0.45824738707424 meters It can be said that the sensor was 45.8 cm away from the center of gravity of the car. The sensor was placed on the dashboard of the car.

Conclusion

The analysis demonstrated the effectiveness of sensor data in vehicle navigation, highlighting the importance of precise calibration and fusion techniques. Magnetometer calibration proved crucial for accurate yaw estimates, while the complementary filter effectively balanced gyro and magnetometer data. Discrepancies between IMU and GPS velocity estimates underscored different sensor limitations. The dead reckoning analysis, integrating IMU data with GPS, offered insights into the vehicle's trajectory, despite some positional errors. Overall, the study reinforces the potential of advanced sensor fusion in enhancing navigation accuracy, crucial for the future of autonomous and assisted driving technologies.